Privacy-Preserving Pneumonia Detection with Federated Learning: Integrating Differential Privacy, Secure Aggregation, Explainability and Uncertainty, and Measuring the Cost of Each Layer

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Abstract—Training chest X-ray classifiers on pooled images is often prohibited by privacy law, and federated learning alone neither prevents leakage from shared updates nor gives clinicians an explanation or a confidence estimate. We build and measure an end-to-end pneumonia detection pipeline that combines federated averaging, sample-level differential privacy (Opacus DP-SGD with an RDP accountant), secure aggregation (Flower SecAgg+), TLS with per-hospital node authentication, Grad-CAM explanations, and Monte Carlo Dropout uncertainty with deferral to human review. A frozen ImageNet DenseNet121 backbone with fixed BatchNorm statistics and a trainable head of about 263 thousand parameters makes per-sample differential privacy valid, shrinks each update to about 1 MB, and keeps secure aggregation affordable. Because every component is established, the contribution is their integration and a seeded measurement of what each layer costs. Scored with the within-hospital (macro) AUROC over three seeds and three simulated hospitals, federation costs 0.017 against centralized training (0.893 to 0.876), secure aggregation costs nothing measurable, and differential privacy at $\varepsilon = 4$ ($\delta = 10^{-5}$) costs a further 0.038 AUROC and 0.065 sensitivity at matched specificity, most of it at the smallest hospital; the full combination reaches 0.830. A pooled-AUROC comparison used earlier overstated the federation cost. No membership-inference attack tried, including a shadow-model attack, separates the private models from chance, so the formal guarantee, with its stated assumptions, is the privacy claim. Auditing the pipeline exposed a defect that made the reported privacy budget a single-round value; the corrected budget was verified in a six-round Docker deployment. We report these results and the limitations explicitly.

Index Terms—Chest X-ray, differential privacy, federated learning, Grad-CAM, pneumonia detection, secure aggregation, uncertainty estimation.

I. INTRODUCTION

Pneumonia is a leading infectious cause of death worldwide, and chest radiography is the first imaging test most patients receive. Convolutional networks can flag pneumonia on frontal films at a level comparable to practising radiologists [1], but they are trained on images pooled in one place, which consent rules and health-data law often forbid. Federated learning keeps images at each hospital and shares only model parameters, which a coordinator averages [2]; it has been argued to be a practical route to multi-institution medical AI [3]–[5].

Federation alone does not make a system private. Parameters computed from patient images carry information about them, and published attacks recover membership, and in some settings the records themselves, from gradients or model outputs [6], [7]. Meaningful protection needs further layers: a formal privacy mechanism on the update, a cryptographic guarantee that the coordinator cannot read any one hospital's update, and an authenticated secure channel. A clinician is also unlikely to act on a bare label; the system should show what drove the prediction and say when it is unsure.

Most pneumonia detectors assume pooled data, and federated studies that add differential privacy (DP) or secure aggregation rarely report what the protection costs in accuracy, calibration, explanation quality, or communication. A hospital deciding whether to adopt such a system needs exactly those numbers. Each technique here is individually well studied, so the novelty is *integration and measurement*, not algorithmic invention. The contributions are:

- A reproducible pipeline that composes federated averaging, sample-level DP with a formal accountant, SecAgg+, TLS with client authentication, Grad-CAM, and Monte Carlo (MC) Dropout with a human-review deferral path, built only from established libraries and with no custom cryptography.
- An architectural choice, freezing the DenseNet121 backbone and federating only a small head, that reconciles BatchNorm with per-sample DP, and a measurement of what that choice costs.
- A seeded ablation ladder (three seeds) scored with a within-hospital protocol that reports the cost of each layer in AUROC, sensitivity, calibration, communication, and time, including results that went against our expectations and defects found and fixed along the way (a privacy-accounting reporting defect, a non-decomposable DP loss).
- An empirical check of privacy (membership inference), explanation quality (Grad-CAM against expert boxes), and label-ambiguity effects on the deployed operating point.

II. RELATED WORK

CheXNet showed that a DenseNet121 [8] can reach radiologist-level pneumonia detection on chest X-rays [1]. Federated learning for medical imaging has been surveyed and demonstrated across institutions [3]–[5], [11], and non-IID data is a known weakness of federated averaging [12]. Differential privacy is formalised in [15]; DP-SGD [16] with a Rényi accountant [17] is available in Opacus [18]. Secure aggregation [19], with its sub-linear-overhead SecAgg+ variant [20], hides individual updates from the server. Grad-CAM [21] provides class-discriminative heatmaps, MC Dropout [22] approximates predictive uncertainty, and calibration is assessed with expected calibration error [23] and conformal sets [24], [25]. To our knowledge, no single published framework combines all of these in one pneumonia pipeline with the per-layer cost reported; that is the gap addressed here.

III. METHOD

A. Pipeline and Threat Model

Three simulated hospitals each preprocess local images (CLAHE [27], resize, ImageNet normalisation, augmentation), train a head on a frozen DenseNet121 backbone with DP-SGD, quantise and mask the update with SecAgg+, and send it over gRPC with TLS and per-hospital client authentication. The server unmasks only the aggregate and applies FedAvg. Fig. 1 shows the data flow; the only artefact that leaves a hospital is a small, DP-noised, masked parameter update.

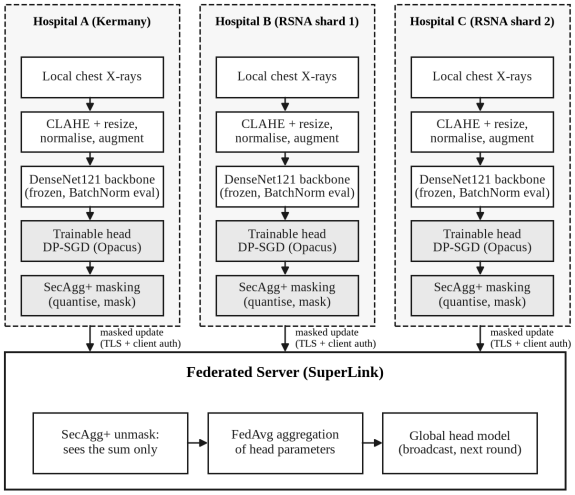


Fig. 1. Architecture of the federated pipeline.

The threat model covers an honest-but-curious server (countered by DP and secure aggregation), a passive network adversary and an unregistered party posing as a hospital (countered by TLS with client authentication), and client collusion up to the SecAgg+ threshold. Malicious or poisoning clients, collusion above the threshold, and side-channel or physical attacks are out of scope. Secure aggregation and Byzantine-update inspection are directly opposed, so robustness to malicious clients is not additive and is left as future work.

B. Frozen Backbone

A stock DenseNet121 relies on BatchNorm, which mixes samples within a batch and therefore breaks per-sample gradients and voids the DP guarantee; averaging BatchNorm statistics across non-IID clients is also unstable. We freeze the ImageNet backbone with BatchNorm in evaluation mode, which makes it a fixed per-sample affine map, and train only a head (Linear $1024 \to 256$, ReLU, Dropout 0.3, Linear $256 \to 2$; about 263 thousand parameters). This simultaneously fixes DP compatibility, reduces each update from about 28 MB to about 1 MB, keeps masking cost low, and fits a 4 GB GPU. Its price is a ceiling on accuracy relative to fine-tuning, which we measure in Section V-H.

C. Differential Privacy

Training uses sample-level DP-SGD: each per-sample gradient g_i is clipped to norm C and Gaussian noise is added to the sum,

$$\tilde{g} = (1/B)\left(\Sigma_i g_i \cdot \min(1, C/\|g_i\|) + N(0, \sigma^2 C^2 I)\right), \quad (1)$$

with σ calibrated from a target ε at $\delta = 10^{-5}$ by the RDP accountant [17], [18]; δ lies well below 1/N for every hospital. We evaluate ε = 1, 2, 4, and 8. The training loss is a per-sample weighted mean, so a sample's gradient does not depend on its batch companions, no class weights derived from label counts are used, and the optimizer settings were chosen on validation data only (Section V-I). Because every hospital adds full noise independently, this is effectively *local* DP, which has worse utility than central or distributed DP at equal ε; ε is reported under that stated trust assumption.

D. Secure Aggregation and Transport

Updates are quantised to integers and masked with Flower's SecAgg+ [14], [20], which uses secret sharing and tolerates dropouts. Masks cancel in the sum, so the aggregate equals plain FedAvg within quantisation error, which a test verifies. Transport uses gRPC with TLS and per-hospital ECDSA key pairs registered with the server, so an unregistered node is rejected; we verified both the rejection and the acceptance paths.

E. Explanation and Uncertainty

Grad-CAM [21] is computed on the final dense block of the global model, client-side. MC Dropout [22] runs T = 20 stochastic passes; the uncertainty is the predictive entropy of the mean distribution. Deferral is by fixed coverage: the highest-entropy 10% of predictions go to a clinician, with the cutoff derived from the validation uncertainty distribution. A chest-X-ray input gate and an isolation-forest check [26] reject or flag out-of-distribution images.

IV. EXPERIMENTAL SETUP

Data. Hospital A is the Kermany chest X-ray set [9] (5,856 images). Hospitals B and C are two shards of the RSNA Pneumonia Detection data [10] (13,342 images each). A size-balanced regime subsamples B and C to Hospital A's size. Splits are at the patient level (RSNA *patientId*; Kermany accession identifiers), with zero overlap across train, validation, and test. The pooled test set has 4,838 images (31.0% pneumonia); pneumonia prevalence is 72% at Hospital A and 22% at B and C. RSNA's native labelling puts both normal films and 'no lung opacity / not normal' films in the negative class; we keep it for benchmark comparability and quantify its consequence in Section V-D.

Protocol. Every run is configuration-driven and seeded (model, data-partition, and client-sampling seeds). Results are mean ± sample standard deviation ($n - 1$) over seeds 42, 123, and 2024, with 20 federated rounds. Model selection, decision thresholds, temperatures, and the deferral cutoff use validation data only, and test labels were read only after every choice was recorded. Because prevalence differs three-fold across hospitals, an AUROC over the pooled test set also rewards telling the hospitals apart. The headline metric is therefore the *macro AUROC*, the mean of the three within-hospital test AUROCs; per-hospital and pooled AUROCs are reported beside it, and sensitivity is measured where each hospital sets a threshold for ≥ 90% specificity on its own validation split and applies it to its own test split. The pipeline runs in simulation (for measurement) and as a Docker Compose deployment with real TLS and node authentication (for demonstration), and is covered by 293 automated tests.

V. RESULTS

A. Ablation Ladder

TABLE I. Macro (within-hospital) test AUROC and sensitivity at a per-hospital 90%-specificity threshold (3 seeds).

Configuration	Macro AUROC	Sensitivity
Centralized (ceiling)	0.893 ± 0.001	0.69 ± 0.01
FedAvg (project recipe)	0.876 ± 0.006	0.64 ± 0.02
FedAvg + SecAgg+ (project)	0.874 ± 0.007	0.64 ± 0.02
FedAvg, DP-matched recipe	0.869 ± 0.005	0.60 ± 0.01
FedAvg + SecAgg+ (DP-matched)	0.865 ± 0.012	0.62 ± 0.04
FedAvg + DP, ε = 8	0.839 ± 0.008	0.56 ± 0.02
FedAvg + DP, ε = 4	0.830 ± 0.010	0.54 ± 0.02
FedAvg + DP, ε = 2	0.817 ± 0.013	0.51 ± 0.01
FedAvg + DP, ε = 1	0.796 ± 0.018	0.48 ± 0.03
Row 6: SecAgg+ + DP, ε = 4	0.830 ± 0.010	0.54 ± 0.02

Table I gives the cost of each layer. Centralized training is the ceiling at 0.893. Federated averaging lowers it to 0.876, a cost of about 0.017; the pooled-AUROC comparison used in earlier analysis suggested nine points because it also rewards between-hospital score offsets. Secure aggregation is not distinguishable from plain federated averaging (0.874 against 0.876 in the original run; 0.865 against 0.869 for the matched recipe through the real Flower SecAgg+ pipeline). Differential privacy at ε = 4 costs a further 0.038 macro AUROC and 0.065 sensitivity against the matched non-private baseline, growing as ε tightens. Row 6, the full combination of federation, secure aggregation, and DP, was run over three seeds through the real pipeline in simulation; its macro AUROC (0.830) equals the DP-only run to within 0.0002 per seed. TLS and node authentication do not alter the numbers and were exercised in the Docker deployment (Section V-I).

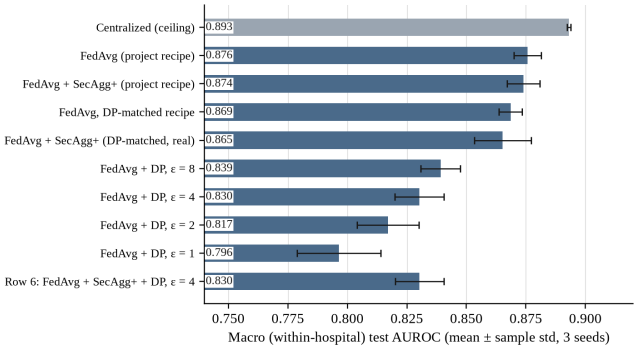


Fig. 2. Macro AUROC across configurations (error bars: sample std).

TABLE II. Every checkpoint on every hospital's own test split (AUROC, mean of 3 seeds).

Model	A	B	C	Pooled
Local A	0.985	0.720	0.738	0.752
Local B	0.723	0.834	0.859	0.799
Local C	0.649	0.834	0.858	0.795
FedAvg	0.946	0.828	0.854	0.814
Centralized	0.979	0.838	0.862	0.905

A like-for-like comparison (Table II) corrects an earlier one that scored local models on their own hospital but the federated model on the pooled set. On the pooled split FedAvg (0.814) is above every local model (0.752–0.799). On each hospital's own data it is within noise of the local model at B and C and about 0.04 below it at A, whose pediatric images differ from the adult RSNA films that dominate training; a local model collapses to 0.65–0.74 on the hospitals it never saw. The value of federation here is that one model serves every hospital at a small in-domain cost without any image leaving its site.

B. Privacy–Utility Trade-off

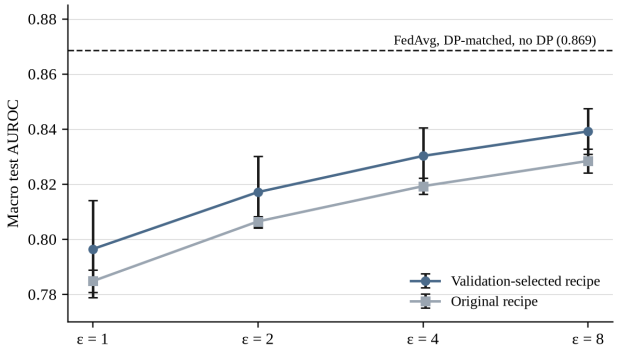


Fig. 3. Privacy–utility curve (macro AUROC, 3 seeds).

TABLE III. DP sweep, validation-selected recipe (mean of 3 seeds).

ε	Macro	A	B	C	Sens.
1	0.796	0.785	0.788	0.817	0.476
2	0.817	0.833	0.794	0.824	0.509
4	0.830	0.865	0.798	0.828	0.538
8	0.839	0.886	0.800	0.831	0.556

Macro AUROC rises monotonically from 0.796 (ε = 1) to 0.839 (ε = 8) (Fig. 3); the curve is shallow because noise is spread over about 263 thousand parameters. The recipe (Adam, learning rate 3×10^{-4}, no class weights) was chosen on validation data after a 48-run sweep; it adds about 0.011 macro AUROC over the original recipe at every ε, about one seed standard deviation. The cost is uneven: Hospital A, the smallest, needs the most noise for a given ε and drops from 0.923 without privacy to 0.865 at ε = 4 and 0.785 at ε = 1 (seed spread 0.03–0.06), while B and C lose 0.03–0.04. The earlier claim that private models miss more than half of pneumonia cases was largely a threshold artefact: at 0.5 the tuned model tracks the class prior (sensitivity 0.19, specificity 0.98), whereas at the per-hospital validation operating point sensitivity is 0.54 at 0.89 specificity. The pooled AUROC can even reverse the comparison (0.847 for DP at ε = 4 against 0.824 for the matched baseline, but macro 0.830 against 0.869).

C. Deployed Operating Point

TABLE IV. Deployed policy, pooled test set (3 seeds; original recipe).

Metric	Value
AUROC / AUPRC	0.7996 ± 0.0121 / 0.6176 ± 0.0247
Sensitivity / Specificity	0.7392 ± 0.0109 / 0.7317 ± 0.0185
Precision / NPV	0.5540 ± 0.0163 / 0.8617 ± 0.0048
F1 / Balanced accuracy	0.6332 ± 0.0106 / 0.7355 ± 0.0087
ECE / Brier (calibrated)	0.0258 ± 0.0007 / 0.1639 ± 0.0048

The deployed model applies a validation-selected threshold (0.35) and temperature scaling (T = 2.2534) to the seed-42 checkpoint of the original recipe, which predates the tuned recipe above (Table IV). It reaches AUROC 0.800 ± 0.012 with sensitivity 0.739 and specificity 0.732, and ECE falls from about 0.10 to 0.026. The AUROC is slightly below the 0.809 of a single deterministic pass because the application averages 20 dropout passes.

D. Label Ambiguity

The 1,761 RSNA images labelled Normal that show other abnormalities account for 753 of the 927 false positives of the deployed seed-42 checkpoint: it flags 42.8% of them, against 9.3% of truly normal RSNA images. Excluding this group at evaluation time raises AUROC from 0.787 (95% CI 0.774–0.800) to 0.884 (0.872–0.896) and specificity from 72.2% to 89.0%, with sensitivity unchanged; this is the same model on an easier subset, not an improvement. Retraining the head-only FedAvg+DP model (ε = 4, 3 seeds) without this group raised clear-label pooled AUROC from 0.907 ± 0.007 to 0.930 ± 0.006 but lowered it on the full test set from 0.809 ± 0.012 to 0.780 ± 0.020, with specificity falling from 0.88 to 0.62, so most of the gap is label ambiguity that retraining does not recover; the original labelling was kept.

E. Calibration, Deferral, and Conformal Coverage

Without DP, MC Dropout confidence is reasonably calibrated (ECE ≈ 0.02–0.03); with the original private recipe ECE is 0.094–0.104 regardless of ε, and temperature scaling repairs most of it. The seed-42 ε = 4 checkpoint has a mean cross-entropy loss of 1.2–2.5 against about 0.5 for non-private FedAvg, suggesting the optimizer settings contributed; the calibration of the tuned recipe was not re-measured. Risk–coverage curves rise monotonically in every configuration: the deferred 10% of cases are classified correctly 48.7% of the time against 75.2% for the rest. Split conformal prediction calibrated for 90% coverage attains 0.908 empirical coverage at ε = 4 (mean set size 1.36).

F. Empirical Privacy

TABLE V. Membership inference: loss attack, its negative control (validation vs test), and a shadow-model LiRA attack (recipe audit; 16–32 shadow models).

Configuration	Loss attack	Control	LiRA	TPR@1%
Centralized	0.5084	0.5038	—	—
FedAvg, no DP	0.4994	0.5033	0.5169	0.0116
DP, ε = 1	0.4950	0.4966	0.4982	0.0098
DP, ε = 4	0.4965	0.4975	0.4992	0.0100
DP, ε = 8	0.4970	0.4978	0.4993	0.0099

A loss attack sits within one point of chance in every configuration, and the highest value, the centralized model's 0.508, is the same size as its negative control (0.504), so it detects nothing. A likelihood-ratio (LiRA) attack [28] with federated shadow models trained on random halves of the data finds a small signal without DP (AUROC 0.517, true-positive rate 1.2% at 1% false-positive rate) and none with DP at ε = 1, 4, and 8 (0.498–0.499). This is evidence against the attacks run, not a bound on leakage: a frozen backbone and a small head memorise little, so the experiment cannot demonstrate the benefit of the privacy layer, and gradient inversion was not attempted. The formal (ε, δ) guarantee is the privacy claim.

G. Explanation Quality

Against RSNA expert bounding boxes, Grad-CAM pointing-game accuracy is 0.184 (centralized), 0.163 (FedAvg), and 0.147 (ε = 4), with mean IoU 0.099, 0.095, and 0.084. Localisation is modest everywhere and federation and privacy lower it only within seed variance; the cause is structural, since a fully frozen backbone gives the classifier only a pooled 1024-number summary.

H. Heterogeneity, Fine-Tuning, and Cost

TABLE VI. Frozen versus fine-tuned last block, no privacy (pooled AUROC).

Configuration	Frozen	Fine-tuned (3 seeds)
Centralized (ceiling)	0.9053	0.9251 ± 0.0004
FedAvg	0.8144	0.8651 ± 0.0101

A Dirichlet sweep isolates heterogeneity: FedAvg reaches a pooled AUROC of 0.776 at α = 0.1 against 0.895 at α = 1.0, a swing larger than the cost of secure aggregation or DP. Unfreezing the last dense block with GroupNorm raises the ceiling and the federated result (Table VI), so the frozen-backbone ceiling is about five pooled-AUROC points in the federated setting; combined with DP at ε = 4 only single-seed indications exist and are not part of the headline.

Plain federated averaging uploads about 1.05 MB per client per round (263 K float32 parameters) whether or not DP is on, against

about 28 MB for a full DenseNet121. SecAgg+, instrumented in the re-evaluation, sends 2.10 MB (the masked update is 64-bit integers, twice the plain update) and receives 4.16 MB per client per round in five messages; key-exchange and unmasking add about 2 KB at three clients, and DP adds no bytes (Flower record bytes, excluding gRPC and TLS framing). A 20-round run with all clients on the GPU takes 83 s with SecAgg+ and 99–109 s with SecAgg+ and DP; Opacus per-sample gradients are about 15 times faster on the GPU than on one CPU core.

I. Verification of the Privacy Accounting

The DP implementation was validated, not only its reported ε. Tests confirm that injected noise has standard deviation σ·C (±2% over 262 thousand coordinates), that removing one sample changes the summed clipped gradient by at most the clip norm, and that the RDP ε is a valid and reasonably tight upper bound (7.08 against an exact analytic 6.57 for the un-subsampled Gaussian mechanism; within 25% of the PRV accountant). Three defects were found. The original weighted loss divided by the batch's total weight, so one sample's gradient depended on its batch companions (a 60% change was measured), violating the sensitivity argument; a per-sample loss is now used. Class weights derived from private label counts lay outside the accountant; the selected recipe uses none. And the SecAgg+ client rebuilt its accountant every round, so the logged ε was the single-round value (2.12 in each of 20 rounds) although the cumulative value is close to 4; the noise had always been calibrated for the full run, so only the reported figure was wrong, in the optimistic direction. The accountant history is now persisted per node; the reported ε rises from 2.12 (round 1) through 3.17 (round 10) to 3.992 (round 20), and an independent recomputation with a fresh accountant gives 3.9955, 3.9939, and 3.9871 for the three hospitals (mean 3.992).

TABLE VII. Docker deployment (SecAgg+ and DP, TLS, 3 registered nodes, 6 rounds, CPU): reported and independently recomputed ε.

Round	Logged ε	Recomputed ε
1	2.943209	2.943209
2	3.230412	3.230412
3	3.455059	3.455059
4	3.649697	3.649697
5	3.829521	3.829521
6	3.991884	3.991884

The fix was tested where it matters: the committed code ran in Docker with three isolated hospital containers, real server-side TLS and registered node keys. Thirty distinct client processes were observed per hospital, equal to six rounds of five secure-aggregation messages, so a fresh process served every message; the reported ε accumulated and equalled the independent recomputation to six decimals (Table VII). This is one seed, on the CPU, with process counts from 1–2 s sampling. The guarantee holds per hospital, per run, at sample level; the many runs of this study, validation-based selection, thresholds, and temperature are not accounted, the subsampled-RDP analysis is taken from Opacus, and its secure random number generator is not enabled.

VI. DISCUSSION AND LIMITATIONS

The principal finding is the *shape* of the cost. Secure aggregation and transport security cost nothing measurable in accuracy; federation costs about 0.017 macro AUROC; and DP adds a further 0.038 together with a 0.065 loss of sensitivity that falls hardest on the smallest hospital. The pooled AUROC exaggerated the cost of federation and could reverse comparisons between configurations, so within-hospital scoring is needed whenever hospitals differ in prevalence. Training only a head keeps the privacy–utility curve shallow and updates small, at the price of an accuracy ceiling of about five pooled-AUROC points.

Limitations. Hospitals are simulated on one machine, with only three clients, so the secure-aggregation guarantee is weak against collusion. DP is effectively local, with worse utility than distributed DP. Ambiguous RSNA labels cap precision. Grad-CAM localisation is weak and the calibration of the tuned private recipe was not measured. Membership-inference evidence cannot separate any recipe from chance. The deployed model is the original-recipe checkpoint of one seed. The DP recipe was tuned once, at ε = 4 on one seed, and reused at other budgets. There is no external validation, the Docker verification is a single six-round CPU run, and malicious participants are not considered. The system is a research prototype: a macro AUROC near 0.83 with an operating point near 73% sensitivity and specificity does not support clinical decisions.

VII. CONCLUSION

We built a pneumonia detector trained across hospitals that never share an image, combining sample-level DP with a formal accountant, secure aggregation, TLS with node authentication, Grad-CAM, and MC Dropout deferral from established components, and measured what each layer costs over three seeds. Scored within hospitals, federation costs about 0.017 AUROC, secure aggregation nothing measurable, and DP at ε = 4 a further 0.038 AUROC and 0.065 sensitivity; the full combination reaches 0.830. Auditing the pipeline corrected the headline comparison and exposed defects in the DP loss, the class weights, and the reported privacy budget, which were fixed and verified, including in a Docker deployment where a fresh process serves every message. Results that went against expectation are reported as found: Grad-CAM localisation is weak and no attack tried separates the privacy layer from chance.

Future work, in order: re-tuning the private recipe for each budget over several seeds and multi-seed runs of the Docker deployment; distributed DP, in which hospitals add shares of the noise; stronger attacks (gradient inversion), external validation, and a larger federation; and defences against malicious participants that reconcile robust aggregation with secure aggregation.

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